$1, 1, \dots, 1$ of length k). Can you find a similar pattern in any real quadratic field $\mathbb{Q}(\sqrt{d})$?

Exercise 3.3.2. A number $u \in (0,1)$ is called *very well approximable* if there is some $\delta > 0$ with the property that there are infinitely many rational numbers $\frac{p}{q}$ with $\gcd(p,q) = 1$ for which

$$\left| u - \frac{p}{q} \right| \leqslant \frac{1}{q^{2+\delta}}.$$

(a) Show that u is very well approximable if and only if there is some $\varepsilon > 0$ with the property that $a_{n+1} \geqslant q_n^{\varepsilon}$ for infinitely many values of n.
(b) Show that for any very well approximable number the convergence in (3.28) fails.

Exercise 3.3.3. Prove Liouville's Theorem[(46)]: if u is a real algebraic number of degree $d \geqslant 2$, then there is some constant $c(u) > 0$ with the property that

$$\frac{c(u)}{q^d} < \left| u - \frac{p}{q} \right|$$

for any rational number $\frac{p}{q}$.

Exercise 3.3.4. Use Liouville's Theorem from Exercise 3.3.3 to show that the number

$$u = \sum_{n=1}^{\infty} 10^{-n!}$$

is transcendental (that is, u is not a zero of any integral polynomial)[(47)].

Exercise 3.3.5. Prove that the theorem of Margulis from p. 6 does not hold for quadratic forms in 2 variables.

3.4 Invertible Extension of the Continued Fraction Map

We are interested in finding a geometrically convenient invertible extension of the non-invertible map T, and in Sect. 9.6 will re-prove the ergodicity of the Gauss measure in that context.

Define a set

$$\overline{Y} = \{(y,z) \in [0,1)^2 \mid 0 \leqslant z \leqslant \frac{1}{1+y}\}$$

(this set is illustrated in Fig. 3.2) and a map $\overline{T} : \overline{Y} \to \overline{Y}$ by

$$\overline{T}(y,z) = (Ty, y(1-yz)).$$

The map $\overline{T}$ will also be called the Gauss map.

Proposition 3.15. *The map $\overline{T} : \overline{Y} \to \overline{Y}$ is an area-preserving bijection off a null set. More precisely, there is a countable union N of lines and curves in $\overline{Y}$ with the property that $T|_{\overline{Y} \smallsetminus N} : \overline{Y} \smallsetminus N \to \overline{Y} \smallsetminus N$ is a bijection preserving the Lebesgue measure.*

PROOF. The derivative of the map $\overline{T}$ is

$$\begin{pmatrix} -\frac{1}{y^2} & 0 \\ 1 - 2yz & -y^2 \end{pmatrix},$$

with determinant 1. It follows that $\overline{T}$ preserves area locally. To see that the map is a bijection, define regions A_n and B_n in $\overline{Y}$ by

$$A_n = \{(y,z) \in \overline{Y} \mid \frac{1}{n+1} < y < \frac{1}{n}\}$$

and

$$B_n = \{(y,z) \in \overline{Y} \mid \frac{1}{n+1+y} < z < \frac{1}{n+y} \text{ and } y > 0\}.$$

These sets are shown in Fig. 3.2. Both

$$\{A_n \mid n = 1, 2, \dots\}$$

and

$$\{B_n \mid n = 1, 2, \dots\}$$

define partitions of $\overline{Y}$ after removing countably many vertical lines (or curves in the case of $\{B_n\}$). Since this is a Lebesgue null set, it is enough to show that $\overline{T}|_{A_n} : A_n \to B_n$ is a bijection for each $n \geqslant 1$, for then

$$\overline{T}|_{\cup_{n\geqslant 1} A_n} : \bigcup_{n\geqslant 1} A_n \longrightarrow \bigcup_{n\geqslant 1} B_n$$

is also a bijection, and we can take for the null set N the set of all images and pre-images of

$$\Big(\overline{Y} \smallsetminus \bigcup_{n\geqslant 1} A_n\Big) \cup \Big(\overline{Y} \smallsetminus \bigcup_{n\geqslant 1} B_n\Big).$$

Notice that $y > 0$ and $0 < z < \frac{1}{1+y}$ implies that

$$0 < yz < \frac{y}{1+y},$$

$$\frac{1}{1+y} < (1 - yz) < 1,$$

and

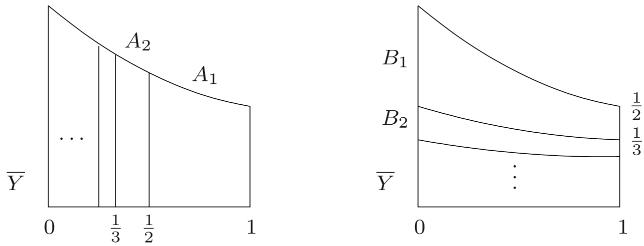

Fig. 3.2 The Gauss map is a bijection between $\overline{Y}$ and $\overline{Y}$, sending the subset $A_n \subseteq \overline{Y}$ to the subset $B_n \subseteq \overline{Y}$ for each $n \geqslant 1$

$$\frac{y}{1+y} < y(1-yz) < y. \tag{3.32}$$

If now $(y,z) \in A_n$ for some $n \geqslant 1$ then $y = \frac{1}{n+y_1}$ for $\overline{T}(y,z) = (y_1, z_1)$ and the inequality (3.32) becomes

$$\frac{1}{n+1+y_1} = \frac{y}{1+y} < z_1 = y(1-yz) < y = \frac{1}{n+y_1},$$

so that $(y_1, z_1) \in B_n$ and therefore $\overline{T}(A_n) \subseteq B_n$. To see that the restriction to A_n is a bijection, fix $(y_1, z_1) \in B_n$. Then $y = \frac{1}{n+y_1}$ is uniquely determined, and the equation $z_1 = y(1-yz)$ then determines z uniquely. Clearly

$$y \in \left(\frac{1}{n+1}, \frac{1}{n}\right)$$

since $y_1 \in (0,1)$, and by reversing the argument above (or by a straightforward calculation) we see that

$$\frac{y}{1+y} = \frac{1}{n+1+y_1} < z_1 < \frac{1}{n+y_1} = y$$

implies $0 < z < \frac{1}{1+y}$ so that $(y,z) \in A_n$. □

Lemma 3.5 gives no indication of where the Gauss measure might have came from. The invertible extension, which preserves Lebesgue measure, gives an alternative proof that the Gauss measure is invariant, and gives one explanation of where it might come from.

SECOND PROOF OF LEMMA 3.5. Let $\pi : \overline{Y} \to Y$ be the projection

$$\pi(y,z) = y \tag{3.33}$$

onto Y. The Gauss measure μ on Y is the measure defined* by

$$\mu(B) = m(\pi^{-1}B)$$

* This construction of μ from m is called the *push-forward* of m by π.

where m is the normalized Lebesgue measure on $\overline{Y}$. Since $\overline{T}:\overline{Y}\to\overline{Y}$ preserves m by Proposition 3.15 and $\pi\circ\overline{T}=T\circ\pi$, the measure μ is T-invariant. □

The projection map $\pi:\overline{Y}\to Y$ defined in (3.33) shows that $\overline{T}$ on $\overline{Y}$ is an invertible extension of the non-invertible map T on Y.

Notes to Chap. 3

(38)(Page 69) The material in Sect. 3.1 may be found in many places; a convenient source for the path followed here using matrices is a note of van der Poorten [294].
(39)(Page 72) In particular, we have Dirichlet's theorem: for any $u\in\mathbb{R}$ and $Q\in\mathbb{N}$, there exists a rational number $\frac{p}{q}$ with $0<q\leqslant Q$ and $|u-\frac{p}{q}|\leqslant\frac{1}{q(Q+1)}$, which can also be seen via the pigeon-hole principle.
(40)(Page 77) A broad overview of continued fractions from an ergodic perspective may be found in the monograph of Iosifescu and Kraaikamp [161]. Kraaikamp and others have suggested ways in which Gauss could have arrived at this measure; see also Keane [187]. Other approaches to the Gauss measure are described in the book of Khinchin [191]. The ergodic approach to continued fractions has a long history. Knopp [205] showed that the Gauss measure is ergodic (in different language); Kuz'min [217] found results on the rate of mixing of the Gauss measure; Doeblin [71] showed ergodicity; Ryll-Nardzewski [326] also showed this (that the Gauss measure is "indecomposable") and used the ergodic theorem to deduce results like (3.26). This had also been shown earlier by Khinchin [190]. Lévy [227] showed (3.25), an implicitly ergodic result, in 1936 (using the language of probability rather than ergodic theory).
(41)(Page 79) These results are indeed easily seen given both the ergodic theorem and the ergodicity of the Gauss map; their original proofs by other methods are not easy. For other results on the continued fraction expansion from the ergodic perspective, see Cornfeld, Fomin and Sinaĭ [60, Chap. 7] and from a number-theoretic perspective, see Khinchin [191]. The limit in (3.26), approximately 2.685, is known as Khinchin's constant; the problem of estimating it numerically is considered by Bailey, Borwein and Crandall [14]. Little is known about its arithmetical properties. The (exponential of the) constant appearing in (3.28) is usually called the Khinchin–Lévy constant. Just as in Example 2.31, it is a quite different problem to exhibit any specific number that satisfies these almost everywhere results: Adler, Keane and Smorodinsky exhibit a normal number for the continued fraction map in [2].
(42)(Page 79) This is proved here directly, using estimates for conditional measures on cylinder sets; see Billingsley [31] for example. We will re-prove it in Proposition 9.25 on p. 323 using a geometrical argument.
(43)(Page 87) Most of this section is devoted to quadratic irrationals, but it is clear there are uncountably many badly approximable numbers; the survey of Shallit [340] describes some of the many settings in which these numbers appear, gives other families of such numbers, and has an extensive bibliography on these numbers (which are also called numbers of *constant type*). For example, Kmošek [203] and Shallit [339] showed that if

$$\sum_{n=0}^{\infty}k^{-2^n}=[a_1^{(k)},a_2^{(k)},\dots],$$

then $\sup_{n\geqslant1}\{a_n^{(2)}\}=6$ and $\sup_{n\geqslant1}\{a_n^{(k)}\}=n+2$ for $k\geqslant3$.

(44)(Page 89) There are many ways to prove this; we follow the argument of Steinig [352] here.

(45)(Page 90) This remarkable uniformity in Definition 3.9 was shown by Woods [388] for $\mathbb{Q}(\sqrt{5})$ and by Wilson [384] in general, who showed that any real quadratic field $\mathbb{Q}(\sqrt{d})$ contains infinitely many numbers of the form $[\overline{a_1, a_2, \dots, a_k}]$ with $1 \leqslant a_n \leqslant M_d$ for all $n \geqslant 1$. McMullen [259] has explained these phenomena in terms of closed geodesics; the connection between continued fractions and closed geodesics will be developed in Chap. 9. Exercise 3.3.1 shows that we may take $M_5 = 4$, and the question is raised in [259] of whether there is a tighter bound allowing M_d to be taken equal to 2 for all d.

(46)(Page 91) Liouville's Theorem [234, 236] (on Diophantine approximation; there are several important results bearing his name) marked the start of an important series of advances in Diophantine approximation, attempting to sharpen the lower bound. These results may be summarized as follows. The statement that for any algebraic number u of degree d there is a constant $c(u)$ so that for all rationals p/q we have $|u - p/q| > c(u)/q^{\lambda(u)}$ holds: for $\lambda(u) = d$ (Liouville 1844); for any $\lambda(u) > \frac{1}{2}d + 1$ (Thue [360], 1909); for any $\lambda(u) > 2\sqrt{d}$ (Siegel [343], 1921); for any $\lambda(u) > \sqrt{2d}$ (Dyson [77], 1947); finally, and definitively, for any $\lambda(u) > 2$ (Roth [319], 1955).

(47)(Page 91) This observation of Liouville [235] dates from 1844 and seems to be the earliest construction of a transcendental number; in 1874 Cantor [47] used set theory to show that the set of algebraic numbers is countable, deducing that there are uncountably many transcendental real numbers (as pointed out by Herstein and Kaplansky [150, p. 238], and despite what is often taught, Cantor's proof can be used to exhibit many explicit transcendental numbers). In a different direction, many important constants were shown to be transcendental. Examples include: e (Hermite [149], 1873); π (Lindemann [232], 1882); α^β for α algebraic and not equal to 0 or 1 and β algebraic and irrational (Gelfond [113] and Schneider [334], 1934).